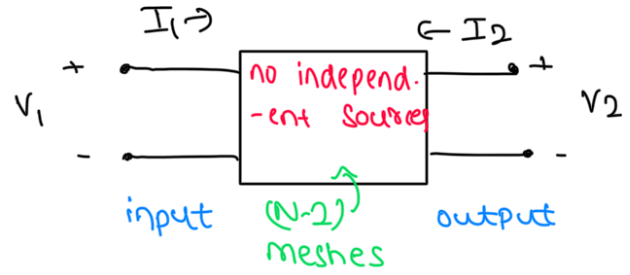
Two port network:

$R = \frac{V}{I}$ in time domain

$Z = \frac{V}{I}$ in laplace domain ($Z \rightarrow$ impedance)

$\frac{1}{Z} \rightarrow$ admitance



$$Z_{11} I_1 + Z_{12} I_2 + \dots + Z_{1N} I_N = V_1$$

$$Z_{21} I_1 + Z_{22} I_2 + \dots + I_{2N} I_N = V_2$$

$$\vdots \qquad = 0$$

$$\vdots$$

$$= 0$$

$$\Rightarrow [Z]_{N \times N} \cdot [I_1 \; I_2 \; \dots]^T_{N \times 1} = [V_1 \; V_2 \; \dots \; 0]^T_{N \times 1}$$

we are only interested in I_1, I_2

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = [Y] \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$[Y]$ ↓ admitance matrix of a two port network

$$[Y] = [[Z]^{-1}]_{\text{restricted to } 2 \times 2}$$

transfer admittance ↑

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$Y_{11} = \frac{I_1}{V_1}\Big|_{V_2 = 0} \qquad Y_{12} = \frac{I_2}{V_2}\Big|_{V_1 = 0}$$

$$Y_{21} = \frac{I_2}{V_1}\Big|_{V_2 = 0} \qquad Y_{22} = \frac{I_2}{}\Big|$$

$V_2|_{V_1=0}$

One port network:

two port network with output one open circuit $(I_2 = 0)$, $(V_2 = 0)$

$$z_{11}I_1 + z_{13}I_2 + \dots + z_{1N}I_N = V_1$$

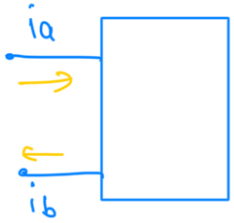
$$z_{21}I_1 + z_{23}I_2 + \dots + I_{2N}I_N = 0$$

$$\vdots \qquad\qquad = 0$$

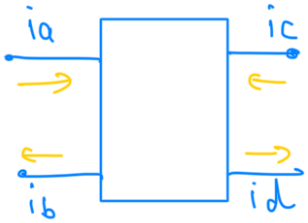
$$\vdots \qquad\qquad = 0$$

$$\Rightarrow [Z_{-1}]\, I = \begin{bmatrix} V_1 \\ 0 \\ \vdots \end{bmatrix}$$

If two port network is Passive (R, L, C) then $Y_{12} = Y_{21}$ (thanks to Reciprocity theorm)

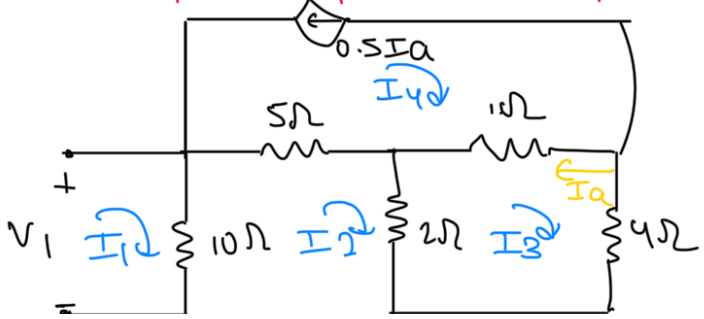


One port network $\Rightarrow i_a = i_b$



Two port network $\Rightarrow i_a = i_b$; $i_c = i_d$

Find the input impedance of the network shown below:



$$10I_1 - 10I_2 = V_1$$

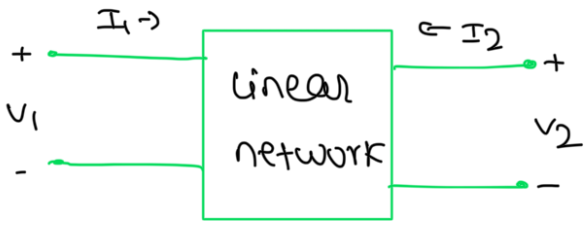
$$-10I_1 + 17I_2 - 2I_3 - 5I_4 = 0$$

$$-2I_2 + 7I_3 - I_4 = 0$$

$$I_4 = 0.5I_a = -0.5(I_4 - I_3) \Rightarrow -0.5I_3 + 1.5I_4 = 0$$

$$\Delta Z = \begin{vmatrix} 10 & -10 & 0 & 0 \\ -10 & 17 & -2 & -5 \\ 0 & -2 & 7 & -1 \\ 0 & 0 & -0.5 & 1.5 \end{vmatrix} = 590\,\Omega^4$$

$$\Delta Z_{11} = \begin{vmatrix} 17 & -2 & -5 \\ -2 & 7 & -1 \\ 0 & -0.5 & 1.5 \end{vmatrix} = 159\,\Omega^3 \Rightarrow Z_{in} = \frac{590}{159} = 3.71\,\Omega$$



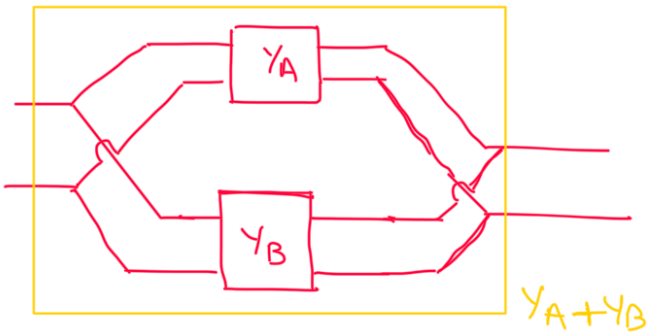
No independent sources

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

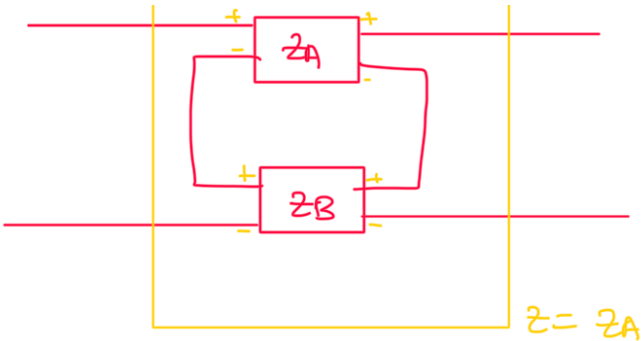
Y → short circuit admitance parameters

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \qquad I = Y \cdot V$$



(parallel)

Same voltage, currents add



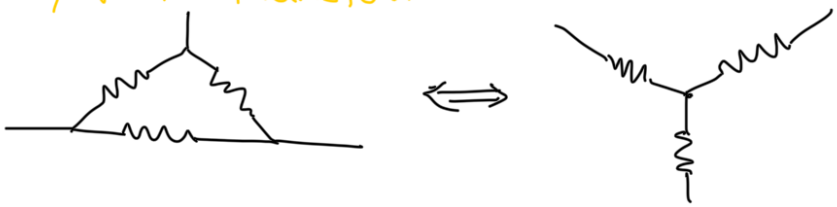
series

same current

voltages add

$Z = Z_A + Z_B$

Π - T / ∇ - Y transform



Other parameters: hybrid (h), transmission (t)

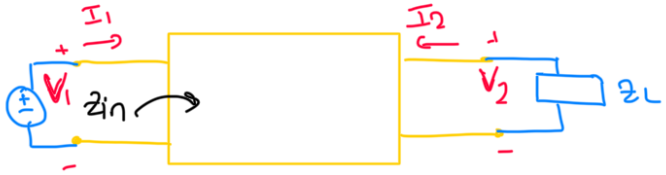
transmission parameters:

Output: $\begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$ Input: $\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

ABCD parameters: $\begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_1 \end{bmatrix}$

parameters in above eq^n are called abcd parameters



$V_1 = Z_{11} I_1 + Z_{12} I_2$

$V_2 = Z_{21} I_1 + Z_{22} I_2$

$V_2 = -I_2 Z_L$

↓ Approach 1

create a circuit equivalent

$Z_{in} = \frac{V_1}{I_1} = Z_{11} + Z_{12} \left[\frac{I_2}{I_1}\right]$

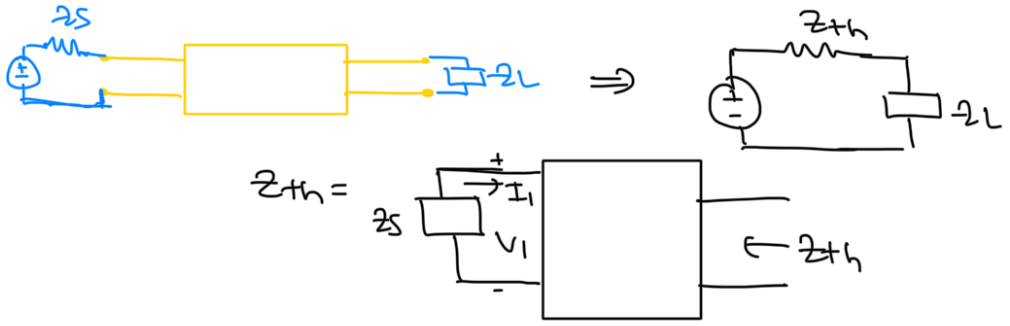
$-I_2 Z_L - Z_{22} I_2 = Z_{21} I_1$

$\rightarrow \frac{I_2}{I_1} = \frac{-Z_{21}}{\quad}$

$$\frac{I_1}{} \quad \frac{}{z_{22}+z_L}$$

$$z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0 \text{ or } z_L=\infty} \qquad \Rightarrow z_{in} = z_{11} - \frac{z_{12}z_{21}}{z_{22}+z_L}$$

$$\frac{V_1}{I_1}\Big|_{V_2=0 \Rightarrow z_L=0} \Rightarrow z_{in} = \frac{z_{11}z_{22} - z_{12}z_{21}}{z_{22}} = \left([\vec{Z}]_{11} \right)^{-1} = [Y_{11}]^{-1}$$



$$-I_1 z_s = V_1 = z_{11} I_1 + z_{12} I_2$$

$$V_2 = z_{21} I_1 + z_{22} I_2 \qquad z_{th} = \frac{V_2}{I_2} = z_{22} - \frac{z_{12}z_{21}}{z_s + z_{11}}$$

$$z_s = 0 \Rightarrow z_{th} = \frac{1}{Y_{22}}$$

---

z parameters:

$$V_1 = z_{11} I_1 + z_{12} I_2$$

$$V_2 = z_{21} I_1 + z_{22} I_2$$

Y parameters:

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

T parameters / ABCD parameters / A'B'C'D' parameters:

$$V_1 = A V_0 - B I_0 \qquad V_1 = A' V_1 - B' I_1$$

$I_1 = CV_2 - DI_2$ $\quad\quad$ $V_2 = C'V_1 - D'I_1$

H parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

G parameters:

$$I_1 = g_{11} V_1 + g_{12} I_2$$

$$V_2 = g_{21} V_1 + g_{22} I_2$$